architectures? How similar are different hidden layer representations to each other? To answer these questions, we use the CKA representation similarity measure outlined in Section 3.1.

We find that as networks become wider and/or deeper, their representations show a characteristic *block structure*: many (almost) consecutive hidden layers that have highly similar representations. By training with reduced dataset size, we pinpoint a connection between block structure and model overparametrization — block structure emerges in models that have large capacity relative to the training dataset.

4.1 Internal Representations and the Block Structure

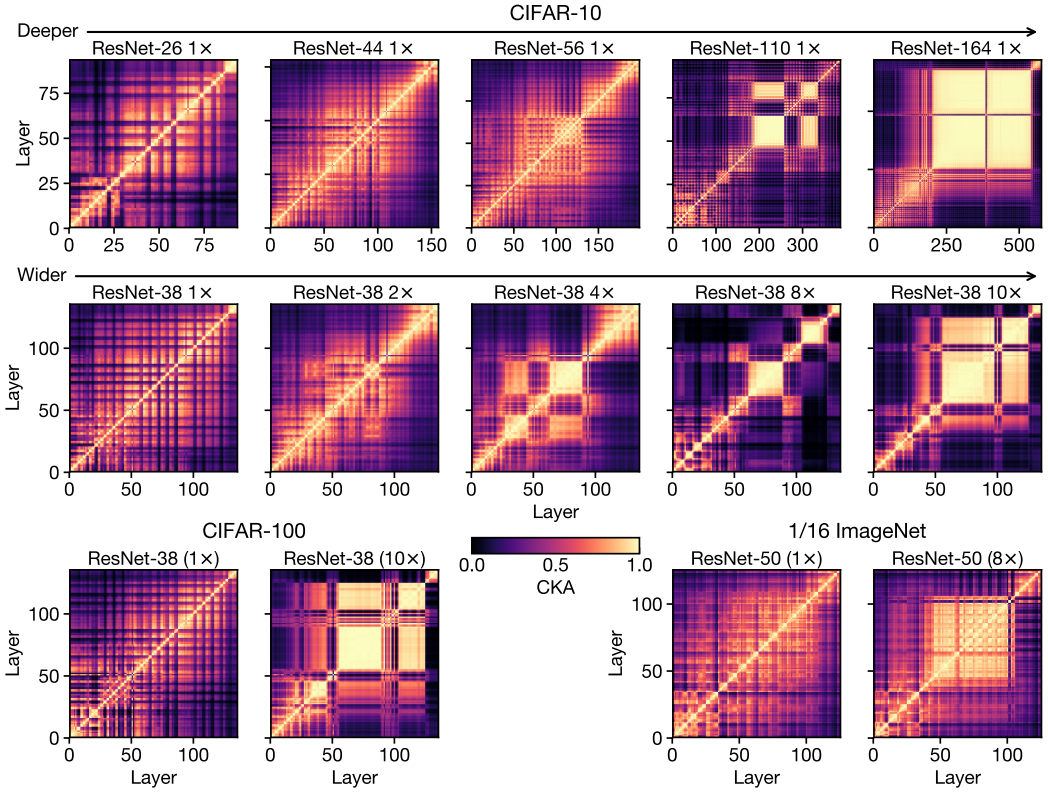

Figure 1: Emergence of the *block structure* with increasing width or depth. As we increase the depth or width of neural networks, we see the emergence of a large, contiguous set of layers with very similar representations — the block structure. Each of the panes of the figure computes the CKA similarity between all pairs of layers in a single neural network and plots this as a heatmap, with x and y axes indexing layers. See Appendix Figure C.1 for block structure in wide networks without residual connections.

In Figure 1, we show the results of training ResNets of varying depths (top row) and widths (bottom row) on CIFAR-10. For each ResNet, we use CKA to compute the representation similarity of all pairs of layers within the same model. Note that the total number of layers is much greater than the stated depth of the ResNet, as the latter only accounts for the convolutional layers in the network but we include *all* intermediate representations. We can visualize the result as a heatmap, with the x and y axes representing the layers of the network, going from the input layer to the output layer.

The heatmaps start off as showing a checkerboard-like representation similarity structure, which arises because representations after residual connections are more similar to other post-residual representations than representations inside ResNet blocks. As the model gets wider or deeper, we see the emergence of a distinctive *block structure* — a considerable range of hidden layers that have very high representation similarity (seen as a yellow square on the heatmap). This block structure mostly appears in the later layers (the last two stages) of the network. We observe similar results in networks without residual connections (Appendix Figure C.1).

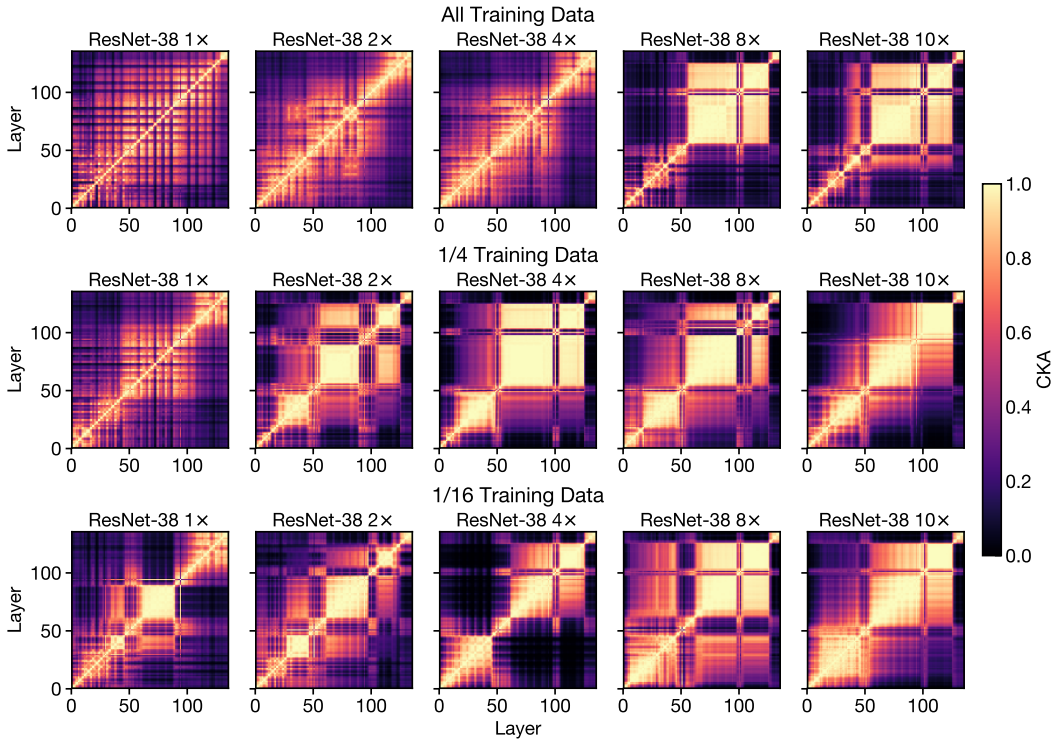

Figure 2: Block structure emerges in narrower networks when trained on less data. We plot CKA similarity heatmaps as we increase network width (going right along each row) and also decrease the dataset size (down each column). As a result of the increased model capacity (with respect to the task) from smaller dataset size, smaller (narrower) models now also exhibit the block structure.

Block structure across random seeds: In Appendix Figure D.1, we plot CKA heatmaps across multiple random seeds of a deep network and a wide network. We observe that while the exact size and position of the block structure can vary, it is present across all training runs.

4.2 The Block Structure and Model Overparametrization

Having observed that the block structure emerges as models get deeper and/or wider (Figure 1), we next study whether block structure is a result of this increase in model capacity — namely, is block structure connected to the *absolute* model size, or to the size of the model *relative* to the size of the training data?

Commonly used neural networks have many more parameters than there are examples in their training sets. However, even within this overparameterized regime, larger networks frequently achieve higher performance on held out data (Zagoruyko & Komodakis, 2016; Tan & Le, 2019). Thus, to explore the connection between *relative* model capacity and the block structure, we fix a model architecture, but *decrease* the training dataset size, which serves to inflate the relative model capacity.

The results of this experiment with varying network widths are shown in Figure 2, while the corresponding plot with varying network depths (which supports the same conclusions) can be found in Appendix Figure D.2. Each column of Figure 2 shows the internal representation structure of a fixed architecture as the amount of training data is reduced, and we can clearly see the emergence of the block structure in narrower (lower capacity) networks as less training data is used. Refer to Figures D.3 and D.4 in the Appendix for a similar set of experiments on CIFAR-100. Together, these observations indicate that block structure in the internal representations arises in models that are heavily overparameterized relative to the training dataset.

5 Probing the Block Structure

In the previous section, we show that wide and/or deep neural networks exhibit a block structure in the CKA heatmaps of their internal representations, and that this block structure arises from the large

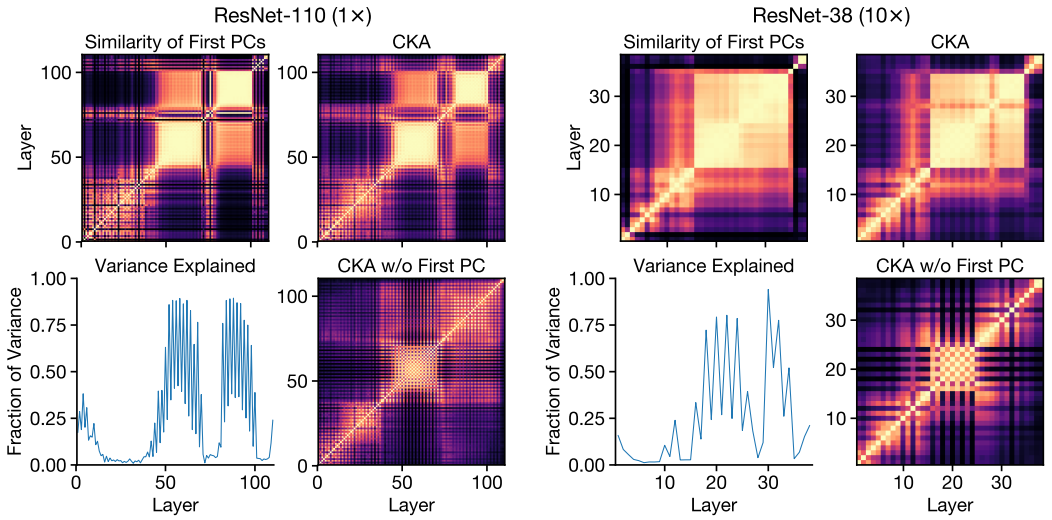

Figure 3: Block structure arises from preserving and propagating the (dominant) first principal component of the layer representations. Above are two sets of four plots, for layers of a deep network (left) and a wide network (right). CKA of the representations (top right), shows block structure in both networks. By comparing this to the variance explained by the top principal component of each layer representation (bottom left), we see that layers in the block structure have a highly dominant first principal component. This principal component is also preserved throughout the block structure, seen by comparing the squared cosine similarity of the first principal component across pairs of layers (top left), to the CKA representation similarity (top right). Compared to the latter, after removing the first principal component from the representations (bottom right), the block structure is highly reduced — the block structure arises from propagating the first principal component.

capacity of the models in relation to the learned task. While this latter result provides some insight into the block structure, there remains a key open question, which this section seeks to answer: what is happening to the neural network representations as they propagate through the block structure?

Through further analysis, we show that the block structure arises from the *preservation* and *propagation* of the first principal component of its constituent layer representations. Additional experiments with linear probes (Alain & Bengio, 2016) further support this conclusion and show that some layers that make up the block structure can be removed with minimal performance loss.

5.1 The Block Structure and The First Principal Component

For centered matrices of activations $\boldsymbol{X} \in \mathbb{R}^{n \times p_1}$, $\boldsymbol{Y} \in \mathbb{R}^{n \times p_2}$, linear CKA may be written as:

$$\mathrm{CKA}(XX^{\mathrm{T}}, YY^{\mathrm{T}}) = \frac{\sum_{i=1}^{p_1} \sum_{j=1}^{p_2} \lambda_X^i \lambda_Y^j \langle \mathbf{u}_X^i, \mathbf{u}_Y^j \rangle^2}{\sqrt{\sum_{i=1}^{p_1} (\lambda_X^i)^2} \sqrt{\sum_{j=1}^{p_2} (\lambda_Y^j)^2}} \tag{4}$$

where $\boldsymbol{u}_X^i \in \mathbb{R}^n$ and $\boldsymbol{u}_Y^i \in \mathbb{R}^n$ are the ith normalized principal components of $\boldsymbol{X}$ and $\boldsymbol{Y}$ and λ_X^i and λ_Y^i are the corresponding squared singular values (Kornblith et al., 2019). As the fraction of the variance explained by the first principal components approaches 1, CKA reflects the squared alignment between these components $\langle \mathbf{u}_X^1, \mathbf{u}_Y^1 \rangle^2$. We find that, in networks with a visible block structure, the first principal component explains a large fraction of the variance, whereas in networks with no visible block structure, it does not (Appendix Figure D.5), suggesting that the block structure reflects the behavior of the first principal component of the representations.

Figure 3 explores this relationship between the block structure and the first principal components of the corresponding layer representations, demonstrated on a deep network (left group) and a wide network (right group). By comparing the variance explained by the first principal component (bottom left) to the location of the block structure (top right) we observe that layers belonging to the block structure have a highly dominant first principal component. Cosine similarity of the first principal components across all pairs of layers (top left) also shows a similarity structure resembling the block structure (top right), further demonstrating that the principal component is preserved throughout the block structure. Finally, removing the first principal component from the representations